Spatiotemporal Correlation Characteristics Between Thermal Infrared Remote Sensing Obtained Surface Thermal Anomalies and Reconstructed 4-D Temperature Fields of Underground Coal Fires

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***Abstract*— Underground coal fires are global catastrophes that result in energy waste, carbon emission, and eco-environment pollution. Remote sensing (RS) detection is essential for underground coal fire extinguishing engineering, and the most used is thermal infrared (TIR) RS. It can well obtain the thermal anomalies of land surface temperature (LST), which is the most direct surface feature of underground coal fires. However, most studies using TIR RS simply delineate underground fire sources vertically according to LST anomalies, which has relatively little impact when initially determining coal fire area locations on the large scale. As for the precise location of small-scale subsurface fire sources, the deviation between subsurface fire source locations inferred and real locations could lead to errors or even mistakes to fire extinguishing engineering. There is a lack of subsurface fire source evolution model reconstruction method, and the spatiotemporal correlations characteristic of LST thermal anomalies and underground fire sources have not yet been discussed. To this end, taking Miquan coalfield (Western China) as an example, a 3-D empirical Bayesian Kriging (EBK3D) method is first proposed to reconstruct the 4-D temperature fields of underground fire sources. Then, the feasibility of the vertical correspondence approach to inferring small-scale subsurface fire sources through LST thermal anomalies detected by unmanned aerial vehicle TIR RS and satellite TIR RS is analyzed. Finally, the spatiotemporal correlation characteristic of LST thermal anomalies and subsurface fire sources is analyzed. As the results show, it is feasible to reconstruct the underground fire source evolution model by the EBK3D method. The reconstructed 4-D temperature fields can dynamically reflect the evolutionary states of underground fire sources in three time periods, with cross-validated root mean square errors of 52.2 °C, 49.6 °C, and 37.1 °C and R^2 of linear regressions of 0.925, 0.9145, and 0.8429, respectively. The LST thermal anomalies show a significant spatiotemporal delay with respect to the subsurface fire source evolution. This makes the locations of the underground fire sources traced by the vertical correspondence method deviate from the real ones. The offsets of underground fire sources relative to surface thermal anomalies in the coal seam strike and dip directions for different time periods at depths of (T1: −44.43 m, T2: −27.72 m, and T3: −20.04 m) are (T1: 73.80 m, T2: 52.33 m, and T3: 45.06 m), and (T1: 16.79 m, T2: 17.27 m, and T3: 24.82 m), respectively. R^2's for the linear regression model of the offset averages in three directions versus time and fire source size are (0.9247, 0.7949, and 0.9564) and (0.8739, 0.85 and 0.9152), respectively.**

***Index Terms*— 3-D empirical Bayesian Kriging (EBK3D), borehole temperature measurement, spatiotemporal correlation characteristics, thermal infrared (TIR) remote sensing (RS), underground coal fire detection, underground fire source.**

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I. Introduction

UNDERGROUND coal fires are phenomena of coal seam burning under natural conditions or caused by human activities [1]. After the underground coal seams or coal seam outcrops encounter oxygen, a certain scale of combustion is formed from low-temperature oxidation and spontaneous combustion to violent combustion with a series of environmental and ecological problems (see Fig. 1). Underground coal fires are global disasters, which are widely distributed in the United States, China, India, Venezuela, Indonesia, South Africa, Australia, and so on [2]. The underground coal combustion constitutes a complex system with thermal, geomagnetic, electric, and geochemical processes involved [3]. Underground coal fires waste massive resources, cause economic loss, and seriously pollute the environment. It also

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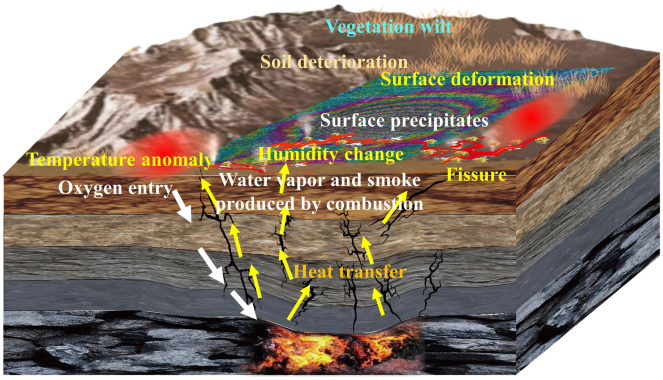


Fig. 1. Series of environmental impacts resulting from underground coal fire combustion.

releases large amounts of greenhouse gases and toxic gases (such as CO_2, CH_4, CO, SO_2, and H_2S), leading to air pollution and exacerbating the global greenhouse effect, further affecting human health and safety.

Coal fire detection is critical for underground coal fire extinguishing engineering. Traditional detection means include isotopic radon measurement [4], index gases [5], self-potential and magnetic [6], temperature measurement [7], and so on. These techniques are highly accurate but inefficient and inapplicable on a large scale. Recently, underground coal fires have been detected with remote sensing (RS) methods because of its low cost, noncontact, time-saving, and wide spatial coverage. These studies are mainly based on optical multispectral [8], [9], thermal infrared (TIR), and radar data, most of which are used for surface feature extraction, thermal analysis [10], [11], [12], and land subsidence accompanying underground coal fires [13], [14]. Besides, some studies combine various RS technologies to monitor coal fire areas [15], [16], [17], [18]. Among the RS coal fire detection techniques, satellite TIR RS (SAT TIR RS) is the one mostly employed, which detects land surface temperature (LST) thermal anomalies that are the most direct surface feature related with underground coal fires.

Coal fire detection by SAT TIR RS is mainly focused on LST retrieval of coal fire areas [19], [20], [21], thermal anomaly feature extraction [22], [23], and dynamic analysis [24], [25], [26] that are related to underground coal fires. SAT TIR RS can detect overall LST thermal anomalies. However, due to its spatial resolution, imaging mode, and topographic effects, it cannot obtain a thorough understanding of surface thermal anomaly behaviors related to underground coal fires. The unmanned aerial vehicle (UAV) TIR RS (UAV TIR RS) has been proven as an effective RS platform due to its flexibility and high accuracy, and has been widely applied in resource investigation and environmental monitoring. Benefited from its high spatiotemporal resolution, UAV TIR RS can characterize the surface thermal anomaly features with more details [27], [28], [29], [30], [31].

However, current research on coal fire detection by TIR RS, especially on underground coal fires, generally assumes that surface thermal anomalies correspond vertically to subsurface fire sources, and accordingly, coal fire areas are circumscribed. This has relatively little impact for initially determining coal fire locations over large areas, and some studies have conducted detection and analysis of the spatiotemporal propagation of coal fires based on the surface thermal anomaly obtained by SAT TIR RS [32], [33], [34]. However, when it is employed for precisely locating small-scale subsurface coal fires, the deviations between the true locations of subsurface fire sources and that inferred by the vertical methods are more significant. This may introduce errors or even mistakes for coal fire extinguishing engineering. There is a lack of reconstruction methods for subsurface fire source evolution model, and the feasibility of using this vertical correspondence approach with SAT TIR RS and UAV TIR RS for fine detection of small-scale subsurface coal fires has not been investigated. Moreover, the spatiotemporal correlation characteristics of the surface anomalies detected by RS and the evolution of subsurface fire sources have not been studied.

This is crucial to infer the locations of underground fire sources through RS means. To this end, the Miquan coalfield with drilling areas, which is one of the most well-known coal fire areas in China and worldwide, is selected as the study area. The main contributions and innovations of this study are threefold. First, a method of subsurface fire source evolution model reconstruction is proposed, and 4-D temperature fields of underground fire sources are reconstructed for the first time. Second, the study proves that the detection capability of SAT TIR RS for small-scale concealed fire sources is limited and prone to omissions. The locations of subsurface fire sources circled by the vertical correspondence method through surface thermal anomalies detected by UAV TIR RS deviate from actual locations. Finally, the spatiotemporal correlation characteristic of LST thermal anomalies and underground fire source evolution is analyzed.

The following of this article is organized as follows. Section II introduces the study area and the dataset used in this study. Section III gives the details of the methods. Results and analysis are presented in Section IV. Discussions are made in Section V. Finally, conclusions are made in Section VI. The overall research framework is shown in Fig. 2.

II. Study Area and Dataset

A. Study Area

The Miquan Sandaoba coalfield, a national key coal fire area, is located in the southeast of Urumqi Midong district, Western China. The coal fire area is a hilly terrain in front of Bogda mountain and has a spatial extent of approximately 482 201 m^2, with elevation varying from 813 to 964 m. The land surface cover is dominated by soils and bare rocks, as well as sparsely distributed pasture and low shrubs. Large areas of collapse have occurred due to mining and underground coal seam burning (see Fig. 3).

The Middle Jurassic Xishanyao Formation (J2x) and the Quaternary (Q) strata are exposed in the coal fire area. The coal seams are in the middle and lower of the J2x, and there are nine layers. The total thickness of the coal seam is 39.59 m, and the total thickness of the minable coal seam is 34.66 m. The coal seams are numbered 39, 41, 42, 43, 45-1, 45-2, 45-3, 45-4, and 45-5. The burned rocks are distributed in coal seams 41∼45, the depth of spontaneous combustion of each coal seam varies, and the regularity is not obvious.

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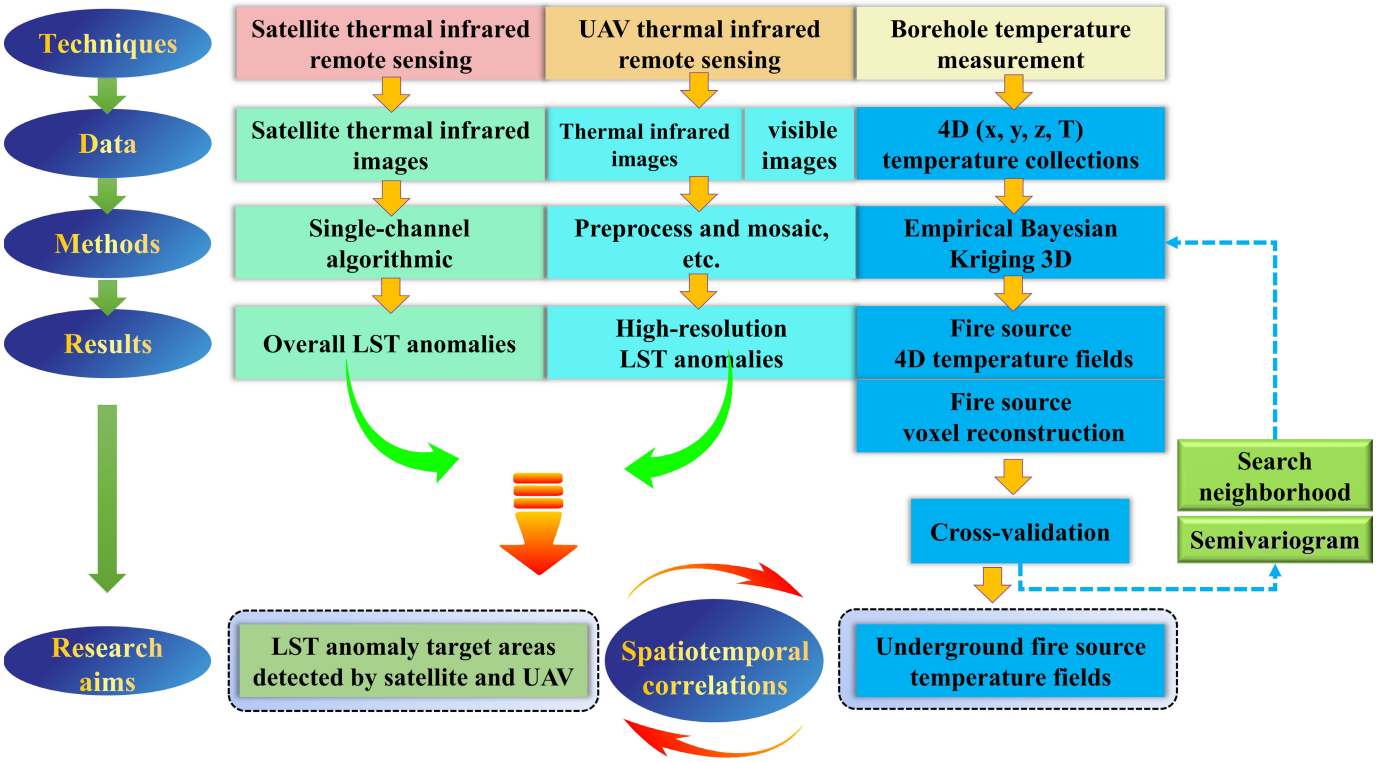


Fig. 2. Research framework.

The fire area is in the south limb of Badaowan syncline, and the basic structural form is a monoclinic structure inclined to the northwest. The stratigraphic dip is 79°∼80°. The shallow part of the fire area burns intensely with complex geological conditions, and the mid-deep part is goaf [35].

The coal seams in this area have the characteristics of high-dipping, extra-thick, and easy spontaneous combustion, so the duration of spontaneous combustion is short. At the same time, years of mining activities in coal mines have caused many goafs, further leading to surface subsidence and formation of fissures and cracks [36]. These cracks and fissures provide the continuous oxygen supply channels for the underground coal fires, further intensifying the coal seam combustion, and the continuous collapse and combustion promote each other to form a vicious cycle.

The fire area develops from west to east with the burning depth gradually becoming shallower from 191 to 20 m (the bottom boundary of the burning is 665 m), with an average of 93 m. A recent field investigation shows that the burning coal seams in the fire area are 39, 41, 42, 43, and 45. The coal seam dip is 327°, and the dip angle is 78°.

B. Remote Sensing Data

The Landsat-8 TIR sensor (TIRS) with two bands denoted as TIR band 10 (in TIRS-1, the bandwidth is 10.60∼11.19 μm) and TIR band 11 (in TIRS-2, the bandwidth is 11.50∼12.51 μm) to measure the TIR radiance at 100 m spatial resolution was used to derive the LST. Considering the climate and meteorological conditions in the study area, as shown in Table I, seven Landsat-8 images with different weather and ground conditions were screened from September 23, 2019, to November 26, 2019.

TABLE I
LANDSAT-8 DATA. THE SENSING TIME IS BEIJING TIME

Date	Path	Row	Sensing time
20190923	142	29	12:50:04
20191009	142	29	12:50:07
20191016	143	29	12:56:19
20191025	142	29	12:50:08
20191101	143	29	12:56:19
20191110	142	29	12:50:07
20191126	142	29	12:50:04

DJI M210 V2 UAV equipped with Zenmuse XT 2 dual-light TIR lens was used to execute three flights of data acquisition in the study area on November 6, 2019. The lens can simultaneously acquire visible and TIR images. The TIR camera has a resolution of 640 × 512, and the field of view is 32° × 26°. The spectral range of the TIR camera is 7.5∼13.5 μm. It has high gain and low gain two modes. The temperature range of high gain mode is −25 °C∼135 °C and −40 °C∼550 °C for the low gain mode. In this study, the TIR camera is set to the low gain mode. The saturation temperature of the UAV TIR camera is 550 °C. The position and attitude of the UAV at the time of the photography are recorded in each image because the format of the UAV TIR image is set to RJPG, which is a storage format specifically for storing temperature information. Acquired TIR images are processed and mosaicked to form a single LST image by the Pix4Dmapper, which is a photogrammetry software, and the LST image represents the absolute temperature. The

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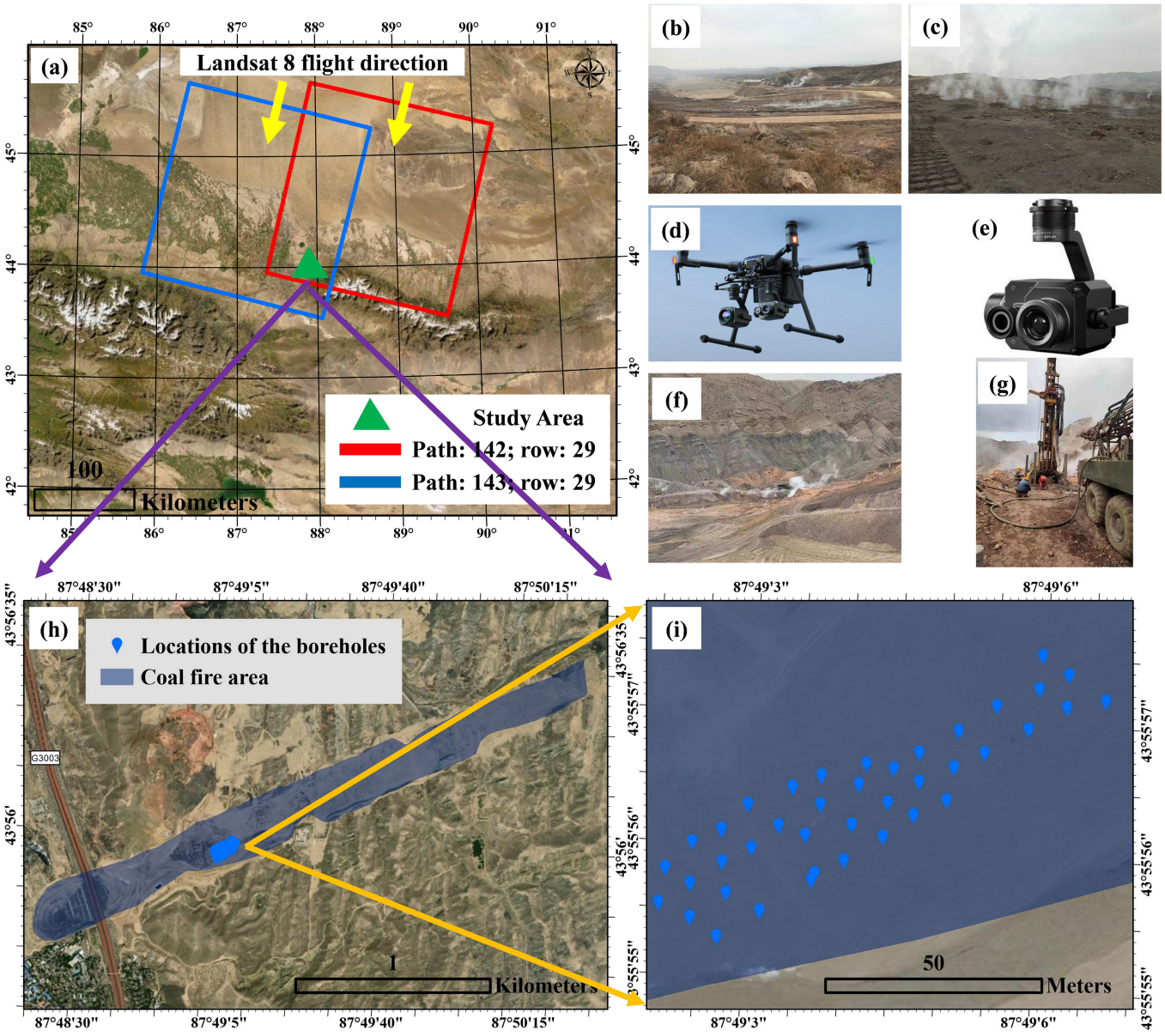


Fig. 3. Research area investigation and measurement campaign. (a) Research area and the coverage of Landsat 8 data. (b) Topography of the study area. (c) Surface released smoke. (d) DJI M210 V2 UAV platform. (e) Zenmuse XT 2 dual-light TIR lens. (f) Small-scale coal fire areas in concealed locations. (g) Field of drilling engineering. (h) Coal fire area. (i) Locations of the boreholes.

surface of the coal fire area is dominated by gravel and sand, and a constant specific emissivity of 0.95 was assumed in this study based on the field investigation. The same method description can be found in [27] and [37]. More details about data acquisition of UAV three flights are shown in Table II.

The flight altitude of the UAV was set at 80 m, and the longitudinal and lateral overlaps were set to 90%. The UAV automatically planned the flight route according to the delineated operation area. After data processing, the visual orthophoto (3.13 cm/pixel) and the LST image (10.90 cm/pixel) of the study area were obtained.

C. Borehole Temperature Measurement Data

Three rounds of borehole temperature data denoted as 20190917∼20190924 (T1), 20190925∼20191003 (T2), and 20191006∼20191014 (T3) were measured during the coal fire monitoring process in a drilling engineering area. The elevations of borehole locations were extracted based on the UAV digital elevation model, and then, the elevations of all underground temperature measurement sampling point (5-m interval in vertical direction) were derived. Furthermore, three datasets of borehole temperature measurements at location (x, y, z) and time (T) were obtained, which were 4-D (x, y, z, T) datasets (see Fig. 4).

III. Methods

A. EBK3D Spatial Interpolation for Borehole Temperature Measurement Data

The empirical Bayesian Kriging (EBK) model is a reliable and fast solution for automatic and interactive interpolation of spatial data. The EBK constructs a robust Kriging model and automatically calculates parameters through subsetting

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TABLE II
UAV DATA. THE DATA COLLECTION TIME IS BEIJING TIME

	Starting time	End time	Number of infrared images	Number of visual images
Flight 1	13:15:30	13:36:08	629	629
Flight 2	13:49:52	14:06:56	522	522
Flight 3	16:51:06	17:14:02	726	726

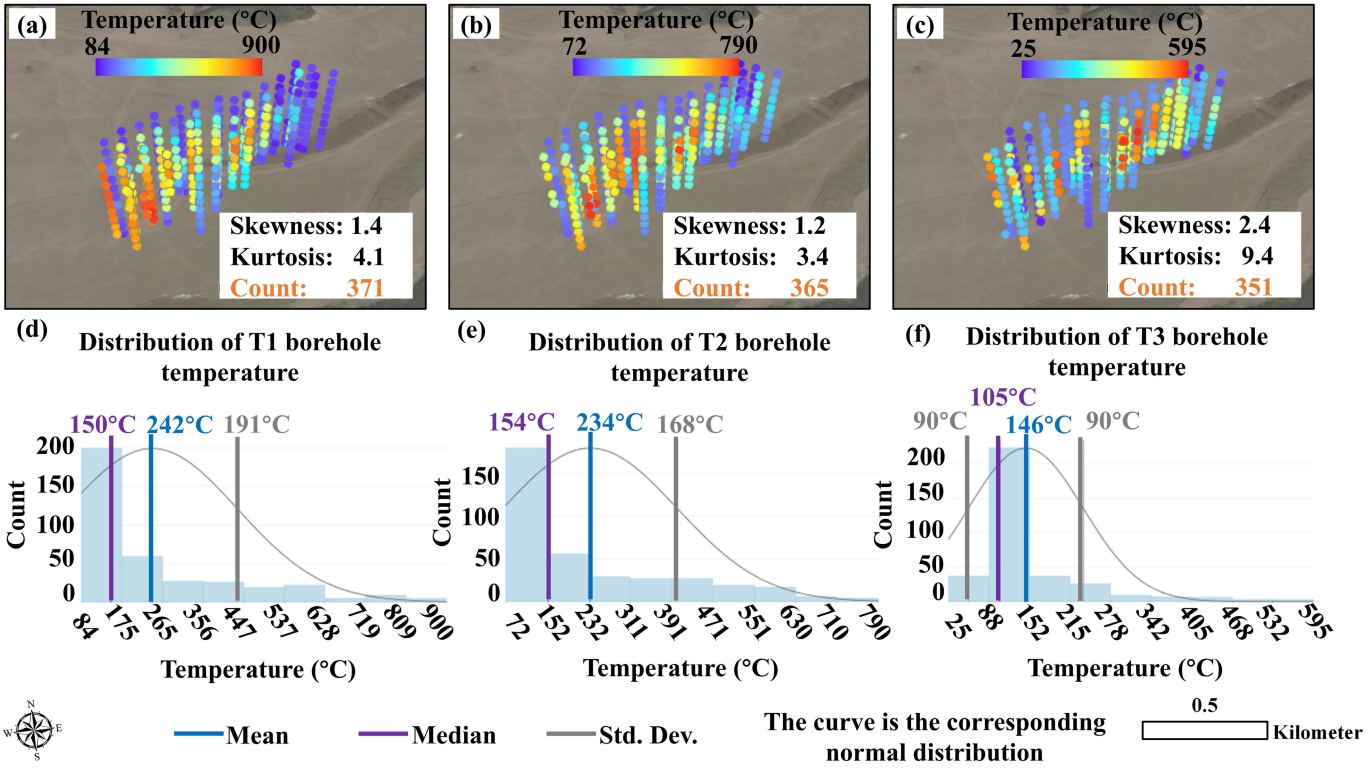


Fig. 4. Distributions of borehole temperature measurement data in different periods. (a)–(c) Spatial distributions of the borehole temperature measurement sampling points for T1, T2, and T3, respectively. (d)–(f) Statistics of the borehole temperature measurements at T1, T2, and T3, respectively. The distribution with a longer right tail has a positive skewness. Distributions with kurtosis larger than 3 have heavier tails and more extreme values than the normal distribution.

at the same time, while other Kriging methods require adjusting these parameters manually for accurate results [38], [39]. Moreover, 3-D EBK (EBK3D) uses EBK to interpolate 3-D sampling points with x and y coordinates, elevation, and the measurements to be interpolated. In addition, EBK differs from traditional Kriging methods by accounting for the error introduced by estimating the semivariogram model [40], therefore helping to reconstruct the underground fire sources precisely through borehole temperature measurement data. The spatial distribution of the underground coal fire was predicted in the following main steps: 1) a semivariogram is estimated from the data; 2) using the estimated semivariogram, a new value is simulated for each position of the input data; and 3) a new semivariogram is estimated using the simulated data. The weight of this semivariogram is then calculated using Bayes' rule [see (1)], which indicates that the likelihood of the observed data can be generated from the semivariogram

$$f_{\text{posterior}}(\Theta|\mathbf{z}) = \frac{f(\mathbf{z}|\Theta)\cdot f_{\text{prior}}(\Theta)}{\int_{\Theta} f(\mathbf{z}|\Theta)\cdot f_{\text{prior}}(\Theta)\cdot d\Theta} \propto f(\mathbf{z}|\Theta)\cdot f_{\text{prior}}(\Theta) \tag{1}$$

where $f_{\text{prior}}(\Theta)$ is the prior distribution; $f_{\text{posterior}}(\Theta|\mathbf{z})$ is the posterior distribution; and $f(\mathbf{z}|\Theta)$ is the conditional probability of the data $\mathbf{z}$, given the model parameters Θ. Each simulated model receives the weight $w_i \propto f(\mathbf{z}|\Theta_\mathbf{i})$, $w_i = f(\mathbf{z}|\Theta_\mathbf{i})/\sum_{i=1}^{K_{\text{sim}}} f(\mathbf{z}|\Theta_\mathbf{i})$, and $\sum_{i=1}^{K_{\text{sim}}} w_i = 1$, where Θ_i are the parameters of the ith prior model simulation. The posterior process is $f_{\text{posterior}}(z(s)|\mathbf{z}) = \sum_{i=1}^{K_{\text{sim}}}(\mathbf{w_i}\cdot f(z(s)|\mathbf{z},\Theta_\mathbf{i}))$, where $f(z(s)|\mathbf{z},\Theta)$ is a conditional distribution of the process $z(s)$ when data $\mathbf{z}$ and model Θ are known. For specific details, please see [41] and [42].

Steps 2 and 3 are executed repeatedly. At each repetition, the estimated semivariogram in Step 1 is used to simulate a new set of values at the input locations. A new semivariogram and its weight are then estimated based on the simulated data. These weights are later leveraged to obtain the predicted values for the unsampled locations.

Various parameters of EBK3D were tried considering the spatiotemporal dynamics characteristic of coal fire and the drastic variation of temperature fields around the underground fire sources. Among them, the semivariogram and the search neighborhood are predominant parameters for getting reliable

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TABLE III
EBK3D MODEL PARAMETERS

Parameters	Values
Semivariogram model type	K-Bessel
Transformation type	log empirical
Subset size	100
Local model area overlap factor	1
Number of simulated semivariograms	100
Order of trend removal	none
Search neighborhood	standard 3D
Max neighbors	1
Min neighbors	1
Sector type	sphere

results. The former defines the weights to determine the contribution of each observation to predict values of unsampled locations. The latter specifies the number and orientation of the neighbors that will be used to predict values at new locations, including sector type, max neighbors, and min neighbors. Although K-Bessel semivariogram is time-consuming, it is chosen because of flexible and accurate features. The search neighborhood should be chosen spherical while ensuring that only one borehole temperature measurement point closest to the point to be interpolated is involved in the calculation to avoid the influence of too many points on the accuracy of the reconstruction results. The remaining parameters are kept as default (see Table III), and we select the best temperature interpolation prediction results of underground fire sources with the minimum value of the root mean square error (RMSE).

To evaluate the reconstructed 4-D temperature fields of underground fire sources, the leave one out cross-validation is used. This method first uses all input points to estimate the parameters of an interpolation model. It then removes a single input point and uses the remaining points to predict the value at the location of the hidden point, and the predicted value is compared to the measured value. The hidden point is then added back to the dataset, and a different point is hidden and predicted. This process repeats through all input points. Then, the RMSE [see (2)] is calculated to evaluate the prediction accuracy and the reliability of the EBK3D model

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(\hat{Z}(s_i) - z(s_i)\right)^2}{n}} \tag{2}$$

where $\hat{Z}(s_i)$ is the predicted value at position s_i, $z(s_i)$ is the measured value at position s_i, and n is the number of sampling points. The above process is performed in ArcGIS Pro 2.8.

B. Single-Channel Algorithmic LST Retrieval

The single-channel (SC) algorithm is used to perform the LST (T_s) retrieval based on Landsat-8 TIRS data using the following general equation [43]:

$$T_s = \gamma\left[\frac{1}{\varepsilon}(\psi_1 L_{\mathrm{sen}} + \psi_2) + \psi_3\right] + \delta \tag{3}$$

where ε is the surface emissivity and (γ, δ) are two parameters given by

$$\gamma \approx \frac{{T_{\mathrm{sen}}}^2}{b_\gamma L_{\mathrm{sen}}}; \quad \delta \approx T_{\mathrm{sen}} - \frac{{T_{\mathrm{sen}}}^2}{b_\gamma} \tag{4}$$

where T_{sen} is the at-sensor brightness temperature; $b_\gamma = c_2/\lambda$ (1324 K for TIRS-1 and 1199 K for TIRS-2; c_2 is the Plank's radiation constants with the value of 14387.7 $\mu\mathrm{m}\cdot K$, and λ is the wavelength); L_{sen} is the at-sensor registered radiance [44]; and ψ_1, ψ_2, and ψ_3 are atmospheric functions, given by

$$\psi_1 = \frac{1}{\tau}; \quad \psi_2 = -L_d - \frac{L_u}{\tau}; \quad \psi_3 = L_d \tag{5}$$

where τ is the atmospheric transmissivity, L_u is the upwelling atmospheric radiance, and L_d is the downwelling atmospheric radiance. These three parameters can be obtained from the following URL (https://atmcorr.gsfc.nasa.gov/).

C. Spatiotemporal Correlation Characteristics Between Underground Fire Sources and LST Thermal Anomalies

The study of spatiotemporal correlation characteristics consists of four steps (see Fig. 5). First, the fire source center locations of T1, T2, and T3 are extracted according to the reconstruction results of underground fire sources in Section III-A. For T1 and T2, the LST thermal anomalies are identified by thermal imaging camera, and the locations are recorded by real-time kinematic. For T3, the centers of the LST anomaly isotherms are extracted from the UAV LST image, and a part of the surface anomaly locations are recorded by field investigation during the UAV data acquisition. Through the above process, the location matrices of the corresponding subsurface fire sources and LST thermal anomalies of T1, T2, and T3 are obtained. Second, the average offset values of the subsurface fire sources relative to the LST thermal anomalies in the eastern, northern, and vertical directions are calculated. Third, the time-series normalization for the acquisition time of the borehole temperature measurement data is carried out. The scale of the underground fire source is calculated based on results of Section III-A. Finally, the evolution of offset values versus time and fire source size is modeled separately to further derive the change rate and change speed of offset values.

IV. RESULTS AND ANALYSIS

A. 4-D Temperature Field Reconstruction of Underground Fire Sources

The temperature fields of underground fire sources reconstructed by the EBK3D method are illustrated in Fig. 6. Generally, the EBK3D-derived results agree with the borehole temperature measurements, as shown in Fig. 4. There is a remarkable fire source with intense burning almost throughout the entire drilling depth appears in the west [see Fig. 6(a1)–(a10)]. However, it can be observed that the range of fire sources at T2 is larger than that of T1 [see Fig. 6(b1)–(b10)]. The fire sources reconstructed at T3 mainly show a clear trend of weakening relative to T1 and T2. It is revealed by Fig. 6(c1)–(c10) that there are almost no fire sources below 25 m, and four fire sources including "one big and three small" are formed in 5∼20 m underground.

There is an obvious fire source forming below 30 m on the west side of T1. It burns intensely around 35∼45 m and starts to weaken below 50 m [see Fig. 6(a7)–(a10)]. Moreover,

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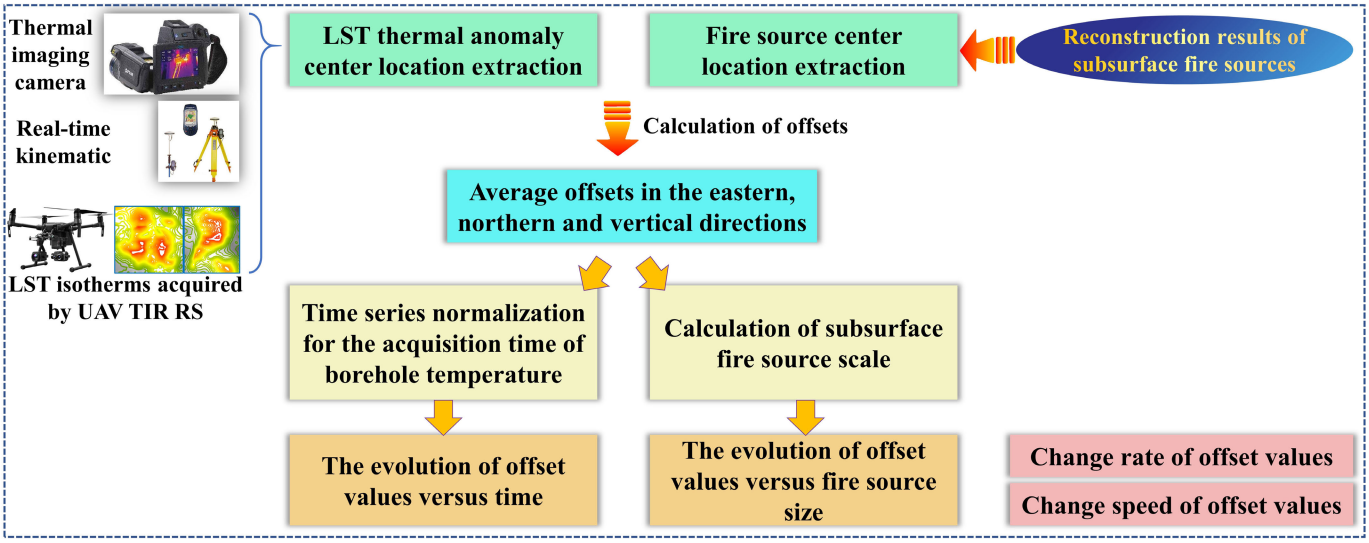


Fig. 5. Study process of spatiotemporal correlation characteristics.

a notable change qualitatively highlights in T1 [see Fig. 6(a8)] and T2 [see Fig. 6(b4)] is that the fire sources move toward the surface and spread to three new fire sources, which burn intensely around 15∼25 m and gradually weaken above 15 m [see Fig. 6(b1)–(b4)]. In particular, the fire source below 25 m near the east is gradually weakened, while the one in the center extends to the northwest and forms a new fire source [see Fig. 6(b5)–(b10)]. When it comes to T3, which is closest in time to the on-site surveying and UAV experiments in the 2019 field campaign, it burns severely at 5∼20 m and forms four obvious fire sources [see Fig. 6(c1)–(c4)]. Among them, the eastern fire source is the largest, and the combustion characteristics below 25 m have gradually disappeared.

This section further focuses on the spatial form and the spatiotemporal evolution process of underground fire sources situated in drilling areas by the 3-D voxel reconstruction of underground space temperature fields (see Fig. 7). The underground fire sources derived from the voxel visualization reveal the weakness process and a trend of moving toward the surface from T1, T2, and T3. For example, the fire source in T1 gradually spreads to the surrounding areas and develops into three smaller fire sources in T2. Furthermore, two fire sources on the west side of T3 gradually diminish, and the intensity of the east side fire source is also weakened compared with T1 and T2 over the horizontal sections made at 35 [see Fig. 7(a4)], 20 [see Fig. 7(b3)], and 15 m [see Fig. 7(c2)], which are the most intense combustion locations that occur in T1, T2, and T3, respectively. Moreover, the ranges of fire sources for the three periods are extracted based on the thresholds of 500 °C and 550 °C.

In order to evaluate the reliability of reconstructed underground temperature fields around fire sources, the method of leave one out cross-validation is exploited. A preliminary validation is conducted in the drilling area via regression analysis using the predicted values for each location as horizontal coordinates and the measured values as vertical coordinates to draw scatter plots. Generally, the points should be uniformly distributed on both sides of the reference line (gray), and the regression line (blue) should coincide with the reference line as much as possible, which means that the prediction of the unknown location of the underground temperature field is unbiased and the model is relatively reliable. Otherwise, the reconstructed results are less stable. It should be noted that the points of three periods are uniformly distributed on both sides of the reference line, and the regression line is in high agreement with the reference line (see Fig. 8), which proves that the reconstructed underground temperature fields are relatively accurate in portraying the temperature variation trend near underground fire sources. The RMSEs between the measured and predicted values of the reconstructed underground temperature fields for three time periods are 52.2 °C, 49.6 °C, and 37.1 °C, respectively. This decreasing trend can be mainly attributed to the more intense combustion at T1 than at T2 and T3. The sudden change of temperature induced by the relatively large temperature difference around fire sources has some influence on the stability of model prediction. With the burning intensity gradually decreasing in T2 and T3, although there are still high-temperature centers, the temperature changes near these are more moderate compared with that in T1; thus, the accuracy of model prediction is improved. In addition, R^2's of linear regressions are 0.925, 0.9145, and 0.8429, respectively.

B. Satellite-Derived LST Anomalies Versus Underground Fire Sources

This section analyzes the detection capability of SAT TIR RS for small-scale concealed fire sources. First, the isotherms near the depths of underground fire sources (35 m for T1, 20 m for T2, and 15 m for T3) are placed on the corresponding satellite detection results of the closest time [see Fig. 9(a)–(c)]. It can be observed from these figures that there are some omitted LST anomaly areas in the satellite-derived results, especially in the case of the corresponding overlying ground surface in drilling areas. This situation can be mainly attributed to the small-scale subsurface fire sources and the topographic effects (see Fig. 10).

Then, the detection results of UAV TIR RS are used as the reference base map, and the technological capacity of SAT TIR RS on how the LST thermal anomalies can be finely detected in the coal fire area is demonstrated. The refined detection

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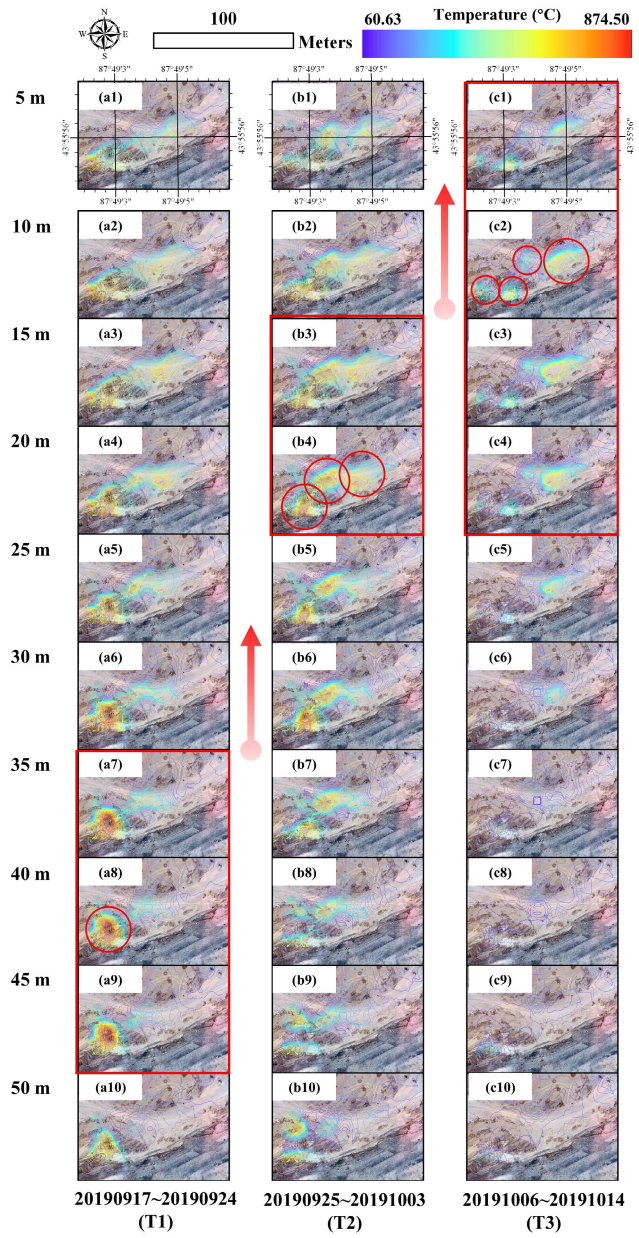


Fig. 6. Isotherms of underground fire sources obtained from EBK3D interpolation for different periods and depths (Relative to the surface, from top to bottom, deeper and deeper). Three columns from left to right represent the underground temperature fields during T1, T2, and T3 respectively. (a1) – (a10) Isotherms of underground fire source at different depths during T1. (b1) – (b10) Isotherms of underground fire source at different depths during T2. (c1) – (c10) Isotherms of underground fire source at different depths during T3.

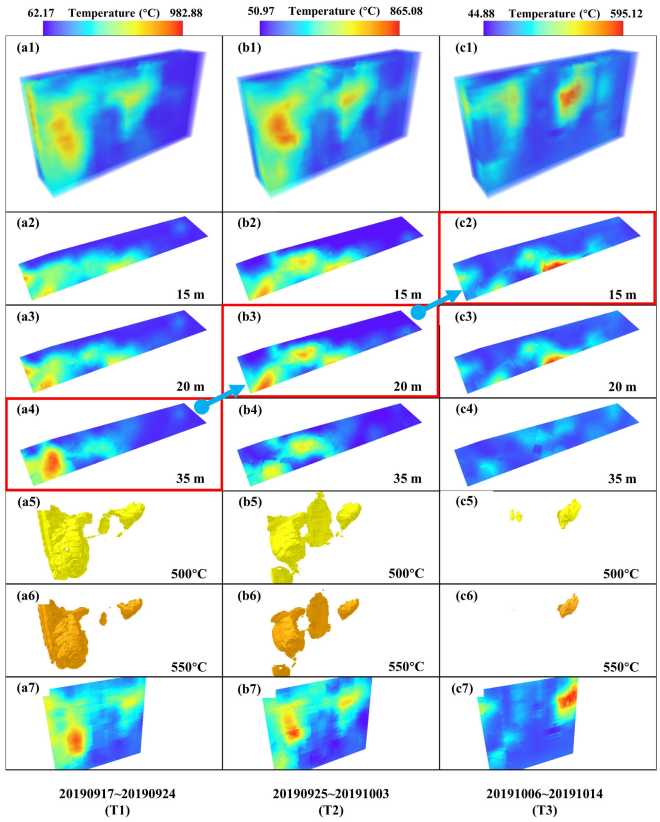


Fig. 7. Voxel visualization of underground fire sources. Three columns from left to right correspond to the voxel visualization results of underground fire sources in T1, T2, and T3, respectively. (a1), (b1), and (c1) are the voxel visualization of underground fire sources in T1, T2, and T3, respectively. Row 2 (a2, b2, and c2), row 3 (a3, b3, and c3), and row 4 (a4, b4 and c4) represent the underground fire sources at different periods and depths, where the most intense burning depths of T1, T2, and T3 are 35 m (a4), 20 m (b3), and 15 m (c2), respectively. Row 5 (a5, b5, and c5) and row 6 (a6, b6, and c6) represent the ranges of underground fire sources extracted with thresholds of 500 °C and 550 °C, respectively. Row 7 (a7, b7, and c7) shows the vertical sections of underground fire sources.

results for SAT TIR RS under different weather and ground conditions are shown in Fig. 10. Overall, it can be seen that the zone with the most severe LST thermal anomalies (R1) and the one with less thermal anomaly intensity (R2) can be effectively detected by SAT TIR RS, except the R3 that can be captured by UAV TIR RS but not SAT TIR RS. A reliable explanation for this exception case might lie in the fact that terrain obscuring is a predominant effect on the missing inspection by SAT TIR RS. Although the depth of underground fire sources is relatively shallow (about 23 m) in R3 and there are also some LST anomaly features formed accordingly, SAT TIR RS with a fixed observation angle is not sensitive to these concealed coal fire sources obscured by terrain.

It is worth remarking that not all coal fire areas present clear thermal anomaly signals in the detection results of SAT TIR RS, like R2. This may be partly because small-scale or new coal fire areas cannot cause sufficient LST thermal anomalies to be apparent in the available SAT TIR RS images. However, $LST_{SAT20191110}$ [see Fig. 10(f)] and $LST_{SAT20191126}$ [see Fig. 10(g)] could more obviously detect the LST anomaly in R2 than the other two images [see Fig. 10(b) and (c)]. Except for the underground coal fire that may have moved and, thus, changed the distribution pattern of LST anomalies, the weather and surface condition might have a significant influence on detection effects. $LST_{SAT20191110}$ and $LST_{SAT20191126}$ are covered by snow, and the lower background ambient temperature and weaker solar radiation make the thermal anomaly information more prominent. The higher signal-to-noise ratio also facilitates the detection of small-scale coal fire areas to some extent.

C. UAV-Derived LST Anomalies Versus Underground Fire Sources

The UAV visual orthophoto, flight paths, and LST thermal anomalies detected by UAV TIR RS are shown in Fig. 11. The black and white signs represent a decreasing

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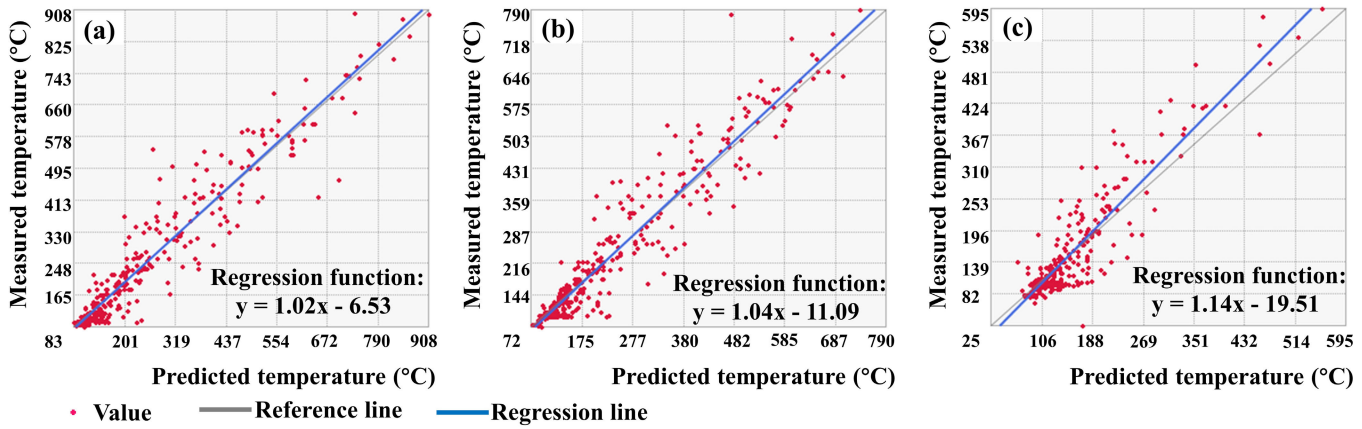


Fig. 8. Cross-validation results of reconstructed underground temperature fields of (a) T1, (b) T2, and (c) T3.

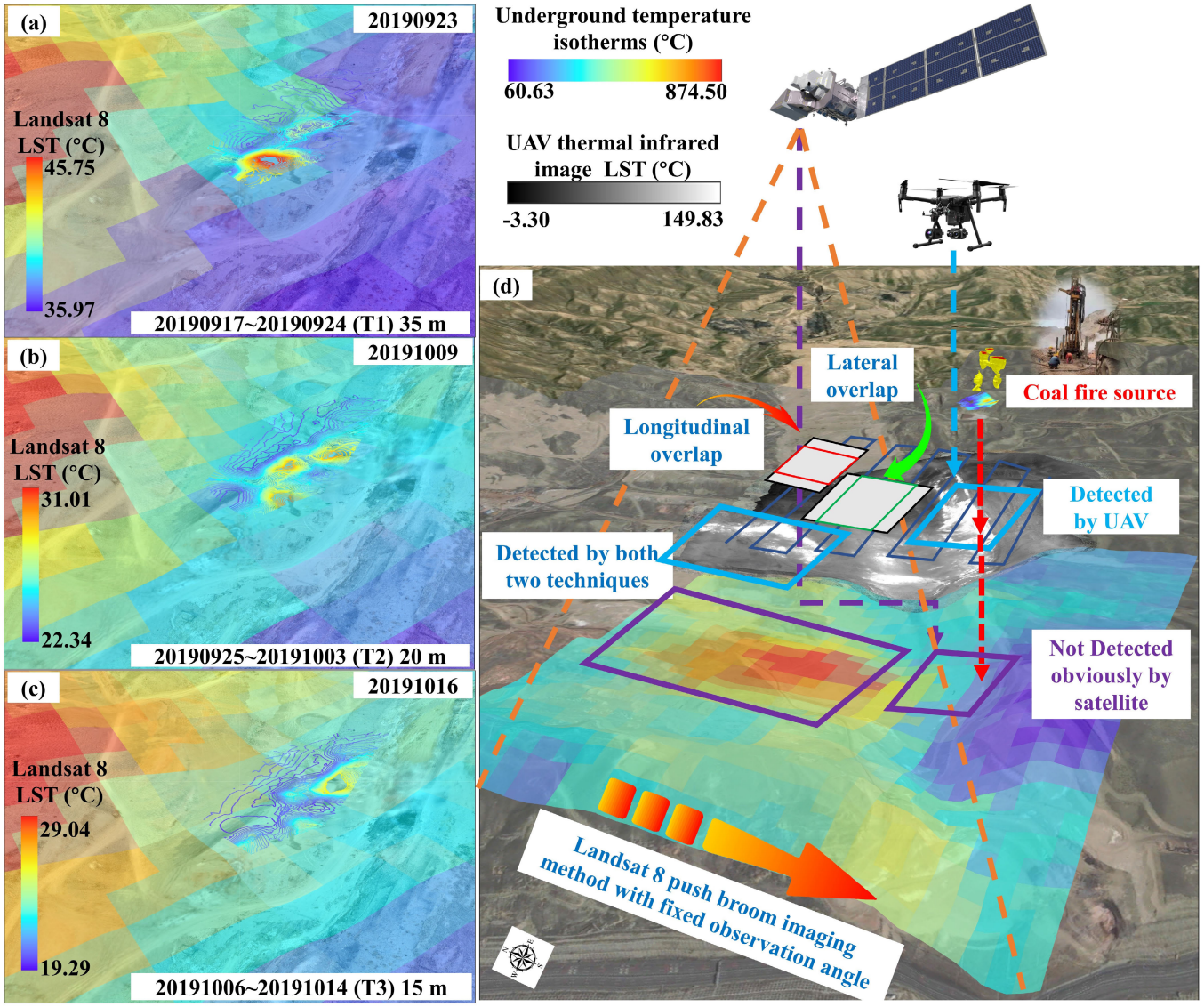


Fig. 9. Underground fire sources with LST anomalies detected by SAT TIR RS and UAV TIR RS. (a)–(c) Closest SAT TIR RS detection results to underground fire source temperature fields of T1, T2, and T3, respectively. (d) Coal fire area detection combined SAT TIR RS, UAV TIR RS, and borehole temperature measurement.

LST and an increasing LST, respectively. It can be observed in Fig. 11(a) that a roughly circular pattern high-temperature zone is detected with a study area of 0.379 km^2. Most of the LST thermal anomalies are related to high-temperature coal fire points, strips, and a certain scale of patches located within the coal seam outcrops and fracture zones, which can

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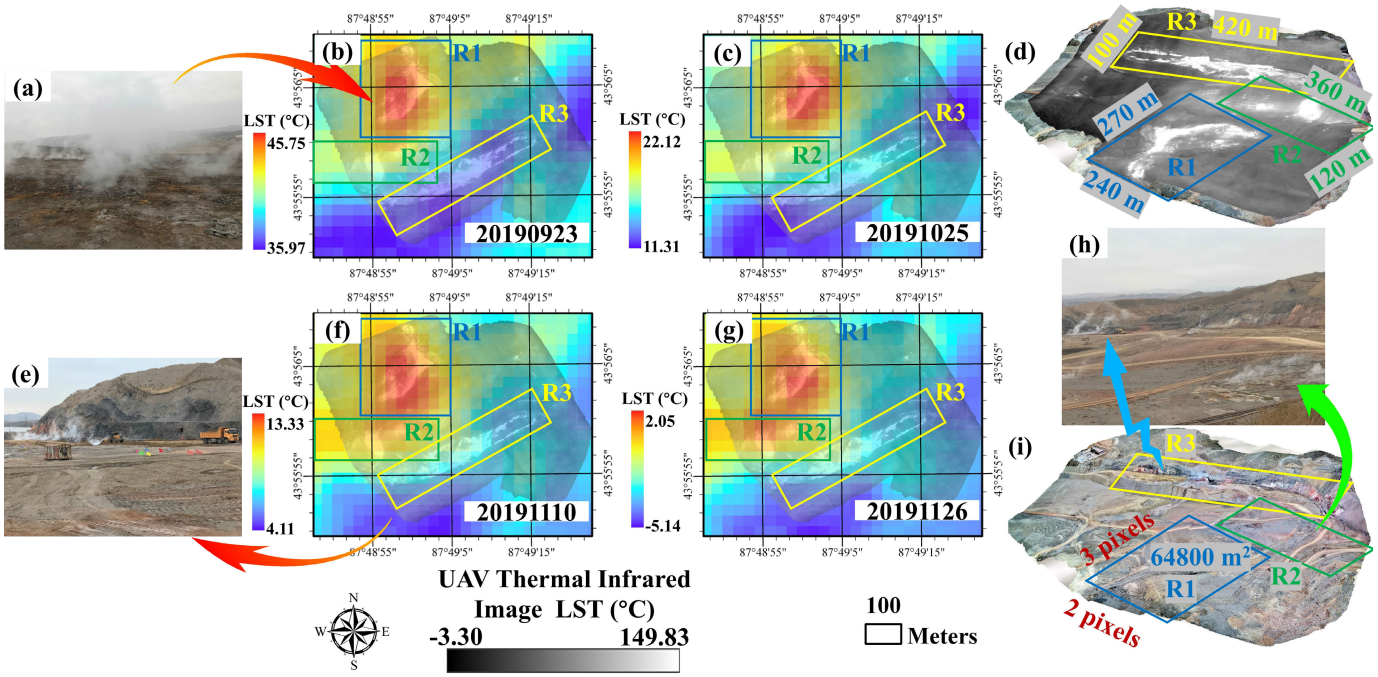


Fig. 10. Refined LST anomaly detection results by SAT TIR RS. (a) Field investigation in the pronounced LST anomaly region R1. (b), (c), (f), and (g) SAT TIR RS detection results under different surface conditions and UAV TIR RS results are used as reference (set 60% transparency) to map three LST anomaly areas, denoted as R1, R2, and R3, respectively. (b) Results of coal fire areas detection when solar radiation is strong, while (c) is the case when it is weak. (d) Range size of R1, R2, and R3 regions. (e) Heat released from the side of the mountain not detected by SAT TIR RS but by UAV TIR RS. (f) Detection results of coal fire areas when the surface is covered with snow, while (g) is the case when the snow cover is strong. (h) Field situation of UAV data collection range. (i) Number of LST anomaly pixels corresponding to the R1 region.

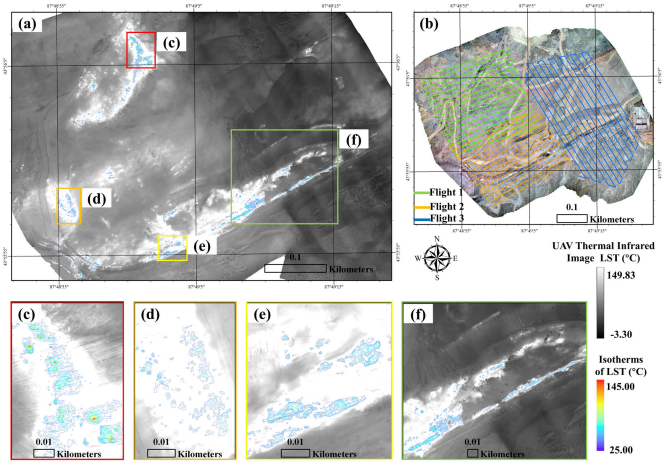


Fig. 11. UAV visual orthophoto and LST images. (a) UAV LST images, with 25 °C as the initial temperature and 5 °C as the interval, to plot isotherms of LST, and four intensely LST anomaly areas, as shown in (c)–(f). (b) UAV visual orthophoto and three flights of UAV.

be portrayed extremely by UAV optical images. It should be noted that the burning of underground coal produces a large amount of smoke, which severely hampers the interpretation of LST thermal anomaly centers located near the fissures from the UAV optical image. Despite this, the UAV-derived isotherm results highlight four LST anomaly zones [see Fig. 11(c)–(f)].

Then, the relative position relationship between LST thermal anomalies detected by UAV TIR RS and subsurface fire sources is analyzed. The fire sources are first extracted from the results of 3-D voxel reconstruction by setting different thresholds (500 °C, 550 °C, 600 °C, and 650 °C). Then, the isosurfaces around underground fire sources at T1, T2, and T3 are obtained. Meanwhile, the LST thermal anomaly areas are generated from the UAV TIR RS images (see Fig. 12). It is also observed from Fig. 12 that there are three obvious LST anomaly areas detected by UAV at the surface overlying drilling area, namely, C1, C2, and C3, which agree with the coal fire data via field inspections and interpretation of UAV optical images. The correlation between LST thermal anomalies and underground fire sources is confirmed by overlying the LST thermal anomaly areas on the subsurface fire sources of three different periods, respectively. It can be inferred that the developments and locations of the underground coal fires do not correspond vertically to that of LST thermal anomalies during the analyzed period. Therefore, there is a bias in using the vertical correspondence method to trap subsurface fire sources by LST thermal anomalies.

It should be remarked that the LST thermal anomaly areas are formed in the southeast of the underground fire sources X1 and X2 [see Fig. 12(d) and (e)]. In particular, the fire sources reconstructed at T3 are characterized by the closest time to the UAV experiments, and field campaigns are selected for further analysis. The correlation between underground fire sources and LST thermal anomalies is demonstrated by quantitative analysis in terms of the horizontal distance (47.37, 37.99, and 26.54 m for C1, C2, and C3, respectively). The slant distances between the center of the fire source Z3 and each high-temperature center detected by UAV are 53.05, 44.49, and 34.72 m for C1, C2, and C3, respectively [see Fig. 12(c)].

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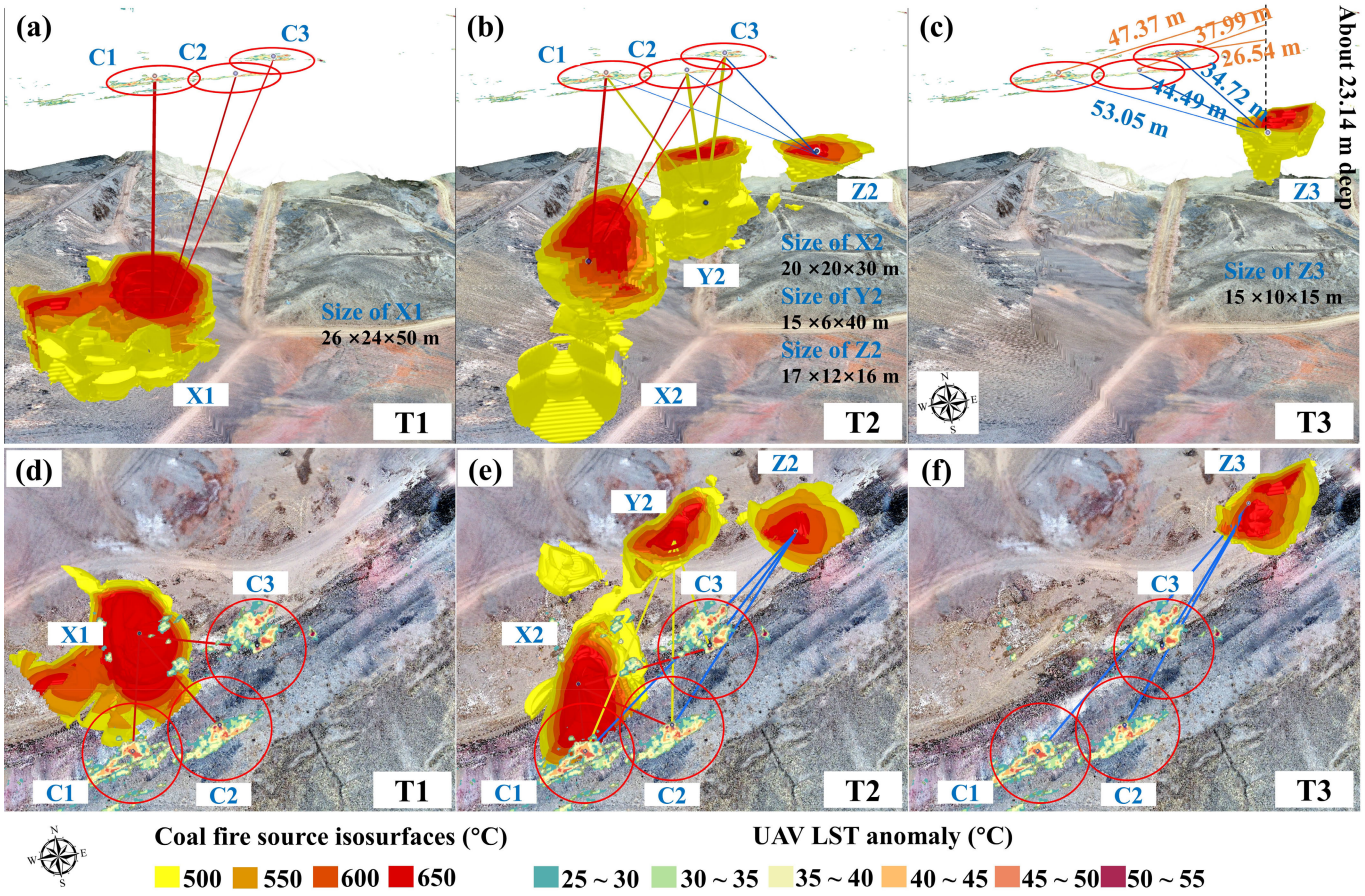


Fig. 12. Underground fire sources and LST thermal anomalies detected by UAV TIR RS (overall vertical offsets are conducted to better show the underground fire sources). (a)–(c) Side views of T1, T2, and T3, respectively. (d)–(f) Top views corresponding to the first row, respectively.

The major evolution trend of the underground fire sources is given as follows. T1 burns intensely, and the fire source X1 is relatively consistent with LST thermal anomalies in the horizontal direction [see Fig. 12(d)]. The fire source X1 gradually moves to the northeast and evolves from one to three. X1 evolves to X2 with a smaller burning range, and the nascent Y2 and Z2 fire sources appear [see Fig. 12(e)]. The location of the fire source Z3 at T3 has a relatively large offset from the UAV-derived LST thermal anomalies [see Fig. 12(f)]. Considering the higher consistency of subsurface fire source at T3 and LST thermal anomalies detected by UAV in terms of data acquisition time, there is a spatiotemporal asynchrony between the evolution of underground coal fire and the surface anomalous features. It may due to the complex mechanism of transmission of heat from underground coal fires to the ground surface.

D. Offset Distances Between Underground Fire Sources and LST Thermal Anomalies

This section analyzes the offset distances of the subsurface fire sources relative to the LST thermal anomalies in three directions (see Fig. 13). The subsurface fire sources have the largest offset relative to the LST thermal anomalies in the eastern direction, and the offset value decreases with time (73.80, 52.33, and 45.06 m for T1, T2, and T3, respectively). The northern offset shows a gradually increasing trend (16.79, 17.27, and 24.82 m), and the offset in the vertical direction indicates that the subsurface fire sources move toward the surface (−44.43, −27.72, and −20.04 m). The standard deviations of the offset distances between the subsurface fire sources and the LST thermal anomalies in three directions at T1, T2, and T3 show that the maximum is in the eastern direction (37.40, 32.93, and 27.23 m) followed by the northern direction (11.33, 11.92, and 10.80 m). The minimum is in the vertical direction (2.56, 2.06, and 1.69 m). These indicating that the LST thermal anomalies caused by the subsurface fire sources are most dispersed in the eastern direction.

V. Discussion

A. Potential and Limits of LST Thermal Anomalies for Accurately Inferring the Location of Underground Coal Fires

It is common knowledge that there exists a straight relationship between underground fire sources and LST thermal anomalies at the macroscale. In addition, the previous study of coal fire has also suggested that subsurface subsidence induced by underground coal fire can lead to systematic orthogonal patterns of opening fissures at the surface. There are correlations between underground fire sources and a variety of surface anomaly features. Thus, for the sake of further improve the knowledge of the coal fire behaviors, a synergistic analysis of LST thermal anomalies mapped from TIR images

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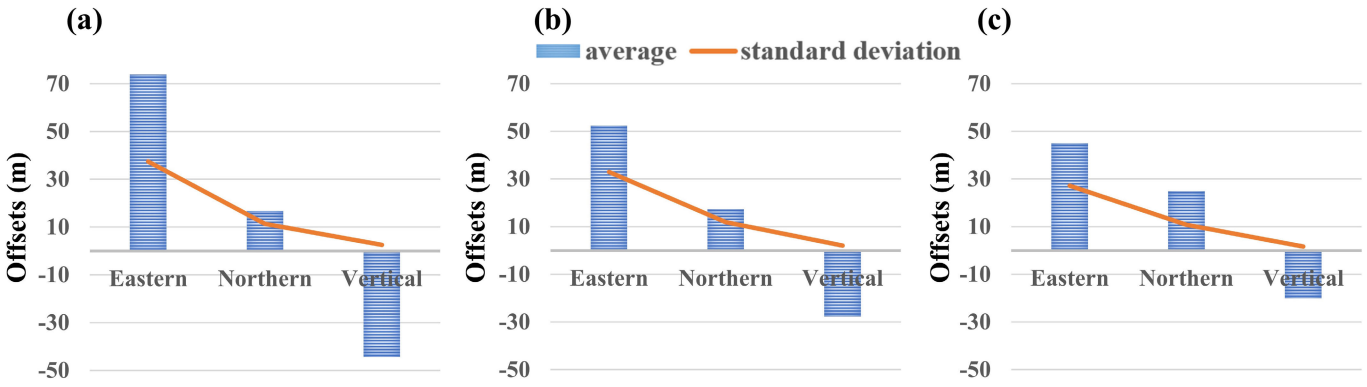


Fig. 13. Offset distances of underground fire sources relative to LST thermal anomalies. (a)–(c) Average and standard deviation of the offset distances between the subsurface fire sources and the LST thermal anomalies in the eastern, northern, and vertical directions for the time periods of T1, T2, and T3, respectively.

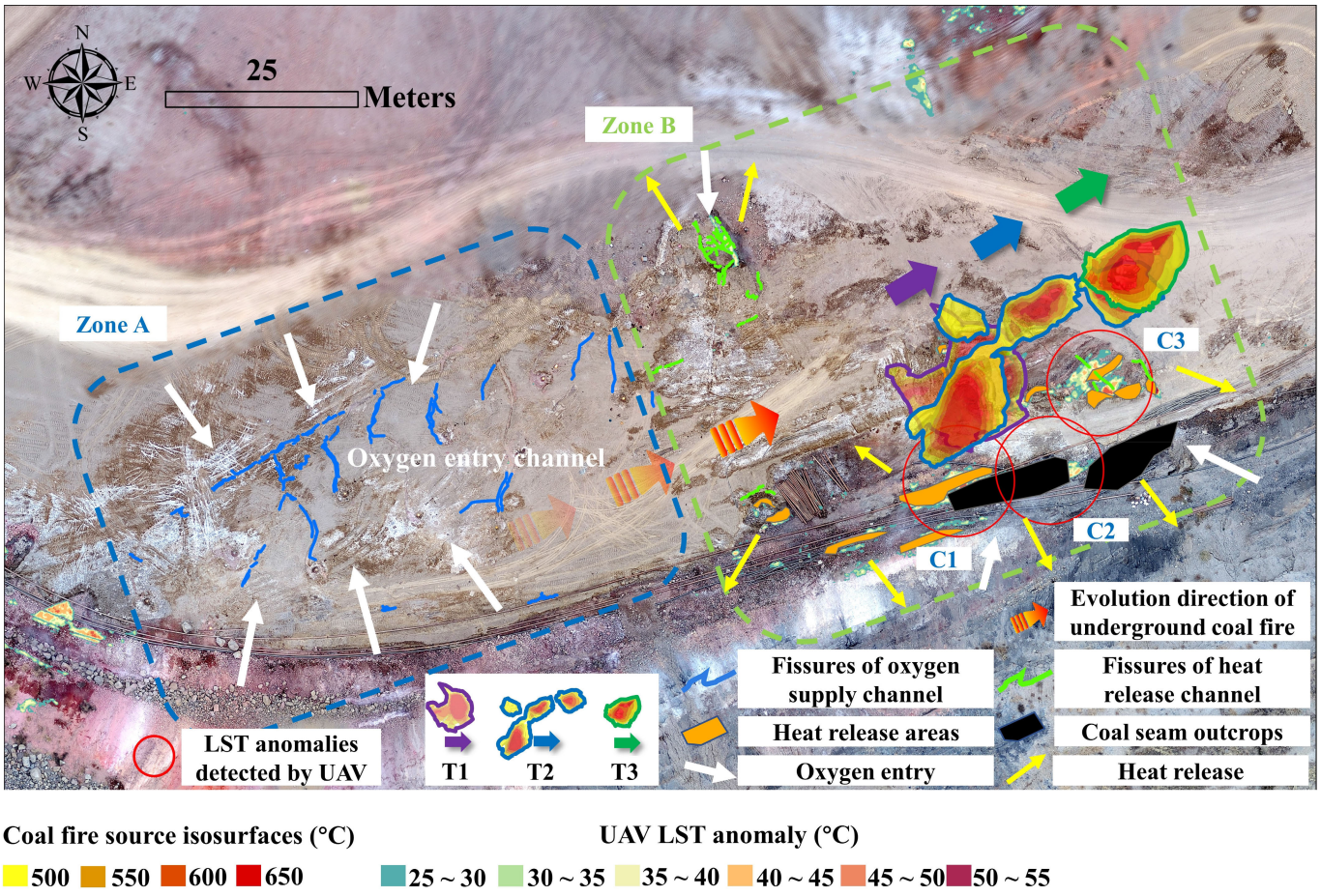


Fig. 14. Spatiotemporal delay characteristics between the evolution of underground fire sources and the abnormal features of ground surface. Zone A is the area of oxygen supply channel that has been formed behind the evolutionary direction of underground coal fire; zone B is the overlying surface of the underground coal fires with a series of surface anomaly features.

and surface opening fissures extracted from high-resolution UAV optical images is conducted (see Fig. 14).

The surface anomaly feature has a spatiotemporal delay relative to the underground fire source evolution. This specifically manifested as the volume of the coal seam changed due to the combustion of the underground coal fires and then resulting in the subsidence of the overlying strata in the Miquan coal fire area. Furthermore, the fractures developed and passed to the ground surface to generate fissures due to the shallow coal fire in the area of interest. Then, stable oxygen supply channels constructed by fissures were formed behind the evolutionary direction of underground coal fires (see the blue dashed rounded rectangle in Fig. 14, noted as zone A). There were no more LST thermal anomalies in zone A since the underground fire source has continued forward along the coal seam inclination to zone B (see the orange dashed arrows in Fig. 14).

The location relationship between LST thermal anomalies and underground fire sources is macroscopically corresponding in the vertical direction. However, there is a deviation between

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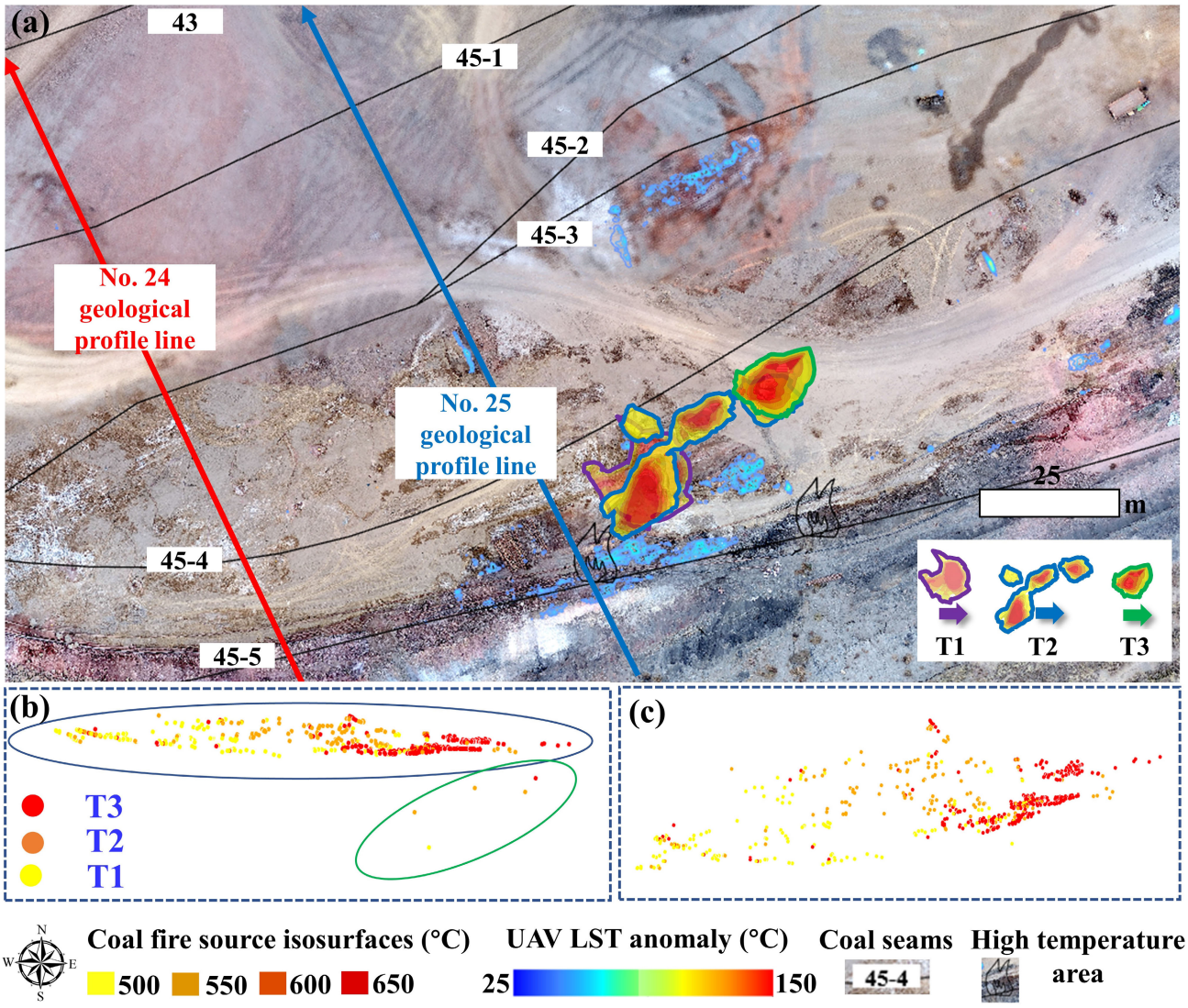


Fig. 15. Distribution of subsurface fire sources and LST thermal anomalies along coal seams and the relative location relationship of LST thermal anomalies to the centers of subsurface fire sources. (a) Distribution of subsurface fire sources reconstructed at T1, T2, and T3 and surface temperature anomalies obtained by UAV TIR RS along coal seams. (b) Side view of the locations of extracted LST thermal anomalies (the blue circle) and subsurface fire sources (the green circle). (c) Top view of the locations of extracted LST thermal anomalies. Yellow, orange, and red points represent the LST thermal anomalies extracted from T1, T2, and T3, respectively.

a particular fire source that burns more intensely underground and its abnormal features at the ground surface. Furthermore, depending on the field investigation and visual interpretation of high-resolution UAV visible images, several features of LST thermal anomaly have been obtained, such as fissures of heat release, heat release areas, and coal seam outcrops, or combinations of them (see zone B in Fig. 14). Zone B in Fig. 14 shows the deviation in spatial horizontal location between the reconstructed underground fire sources and the ground surface anomaly features. The generated heat is generally released outward following the surface fissures and coal seam outcrops around the vertically corresponding surface position of the underground fire sources. The underground fire sources reconstructed from the borehole temperature measurement data for the three time periods indicated by the different colored external lines and transparency are roughly along the northeast direction, which are shown by the purple, blue, and green arrows in Fig. 14. There is a significant spatiotemporal delay in the development of surface anomaly features relative to the movement of the underground fire sources from the location relationship between the underground fire source of T3 and the LST anomalies detected by UAV, as well as other surface anomaly characteristics induced by underground coal fire (see zone B in Fig. 14). Accordingly, this provides some references and insights that the locations of underground fire sources should be reasonably corrected by combining the transport mechanism of it when inferring the real-time state of the underground fire source locations from the surface anomalous features detected by RS.

B. Verification of Spatiotemporal Delay Characteristics of LST Thermal Anomalies Relative to the Subsurface Fire Source Evolution

The underground coal seam distribution and geological profiles are combined to further verify the spatiotemporal delay characteristics of LST thermal anomalies relative to the subsurface fire source evolution. The offsets of both along the coal seam strike and dip are consistent with the coal seam

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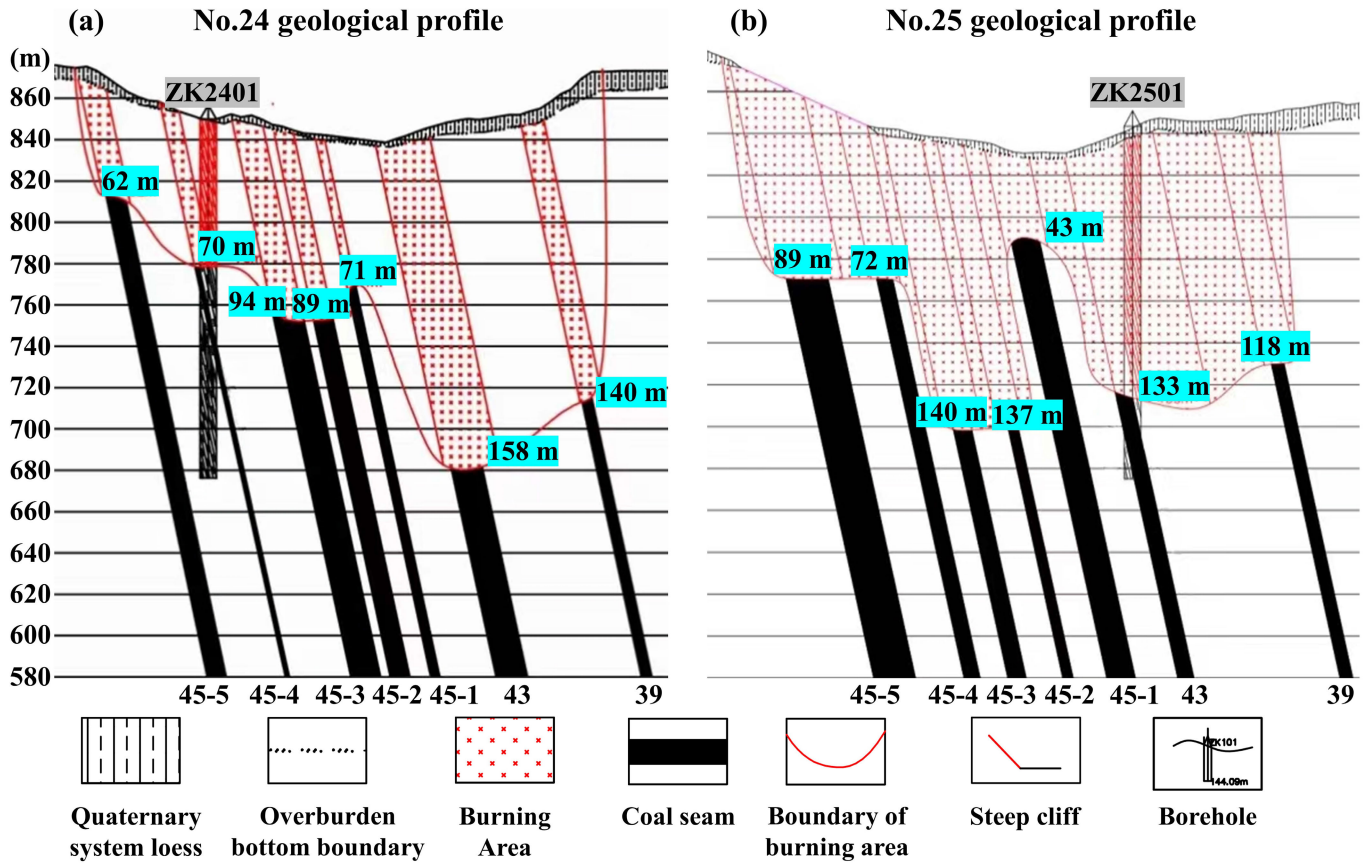


Fig. 16. Geological profile in the study area. (a) No. 24 geological profile. (b) No. 25 geological profile.

distribution, which is roughly along the west–east direction and slightly offset to the north [see Fig. 15(a)]. The offsets of subsurface fire sources and LST thermal anomalies in the coal seam strike gradually decrease with time (T1: 73.80 m, T2: 52.33 m, and T3: 45.06 m). Therefore, in order to infer the subsurface fire source locations more accurately through LST thermal anomalies, the time effect on the offset distance needs to be considered. Specifically, the subsurface fire source burns most intensely at T1. Since underground coal seams have not been burned extensively, sufficient coal and the proximity to oxygen supply channels allow underground coal fires to develop rapidly along coal seams. Thus, before the surface thermal anomaly with a certain scale is formed directly above the underground fire source, it has continued to move and develop along the coal seam strike, while the heat is still released from original channel. Consequently, the offsets between LST thermal anomalies and subsurface fire sources are relatively large during the period when underground coal fires are burning intensely. If underground coal fires can be controlled at the early stage, the effect of the extinguishing engineering is obvious, and the waste of resources can be minimized. The grouting fire extinguishing technology is usually adopted to control coal fires at the early stage. Considering offsets between subsurface fire sources and LST thermal anomalies along the coal seam strike may be large at this stage, there should be a relatively large offset distance for the selection of grouting fire extinguishing locations.

The combustion intensity and scale of subsurface fire sources decrease at T2 and T3, which make offsets of subsurface fire sources relative to LST thermal anomalies along coal seam strike reduce compared with T1 by 29.09% and 38.94%, respectively. In addition to the reduction of coal, the fire sources are far away from oxygen supply channels resulting in insufficient oxygen supply, and new oxygen supply channels have not been formed, which slows down the movement speed of subsurface fire sources. The above analysis shows that offsets of subsurface fire sources and LST thermal anomalies along coal seam strike may be common in each stage of the underground coal fire evolution. The influence of offsets on the delineation of subsurface fire sources and the selection of grouting fire extinguishing locations should be fully considered. When UAV TIR RS is used to obtain high-resolution LST images for coal fire extinguishing engineering, the subsurface fire source location should be corrected by different offset distances according to the time stage of coal fire evolution to locate underground fire sources more accurately.

The offset of the subsurface fire source relative to the LST thermal anomaly in north direction (coal seam dip) is smaller than that in east direction (coal seam strike), with 16.79, 17.27, and 24.82 m for T1, T2, and T3, respectively. The offset of both along north direction increases with time, which is different from that in east direction. The geological profiles (see Fig. 16) show that steep coal seams tilt northward, but the thickness of the underground coal seam makes the offset

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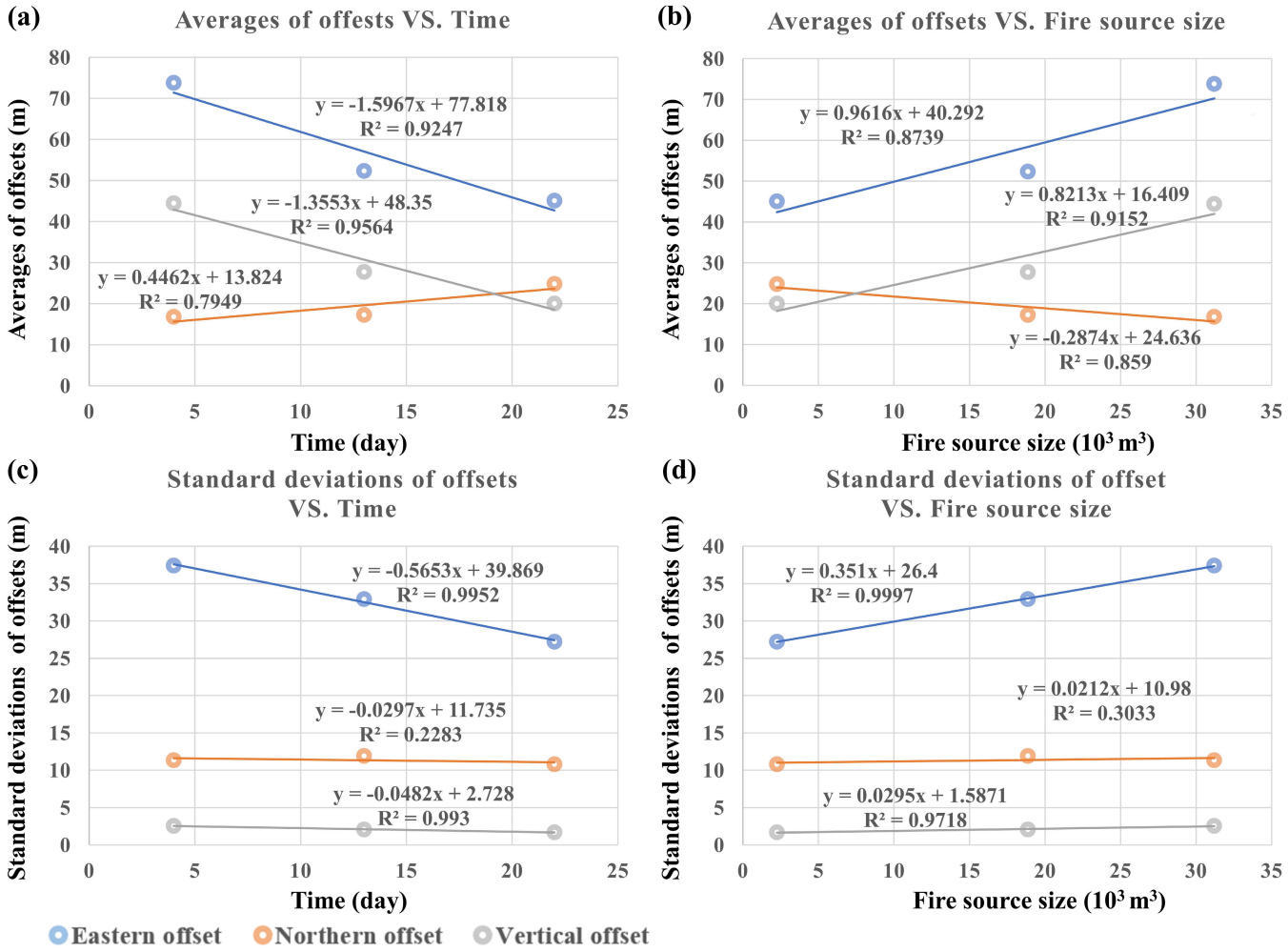


Fig. 17. Averages and standard deviations of offsets of subsurface fire sources relative to surface thermal anomalies versus time and fire source size. (a) and (b) Averages of offsets between the subsurface fire sources and surface thermal anomalies versus time and fire source size, respectively. (c) and (d) Standard deviations of offsets between the subsurface fire sources and surface thermal anomalies versus time and fire source size, respectively.

along coal seam dip limited (about 20 m). Fig. 15(a) shows that the subsurface fire sources are roughly between coal seams 45-5 and 45-4, and coal seam 45-4 is thin and distant from 45-3, so it cannot provide combustion material conditions for the continued development of underground coal fires to the north. The analysis of reconstructed subsurface fire sources shows that the combustion is relatively intense at T1, and offsets of north direction increased by 2.86% and 47.83% for T2 and T3 compared with T1. The less percentage increase from T1 to T2 is mainly due to that there are three obvious fire sources for T2, and these three fire sources make the overall combustion scale larger. Thus, after averaging locations of subsurface fire sources, the offset is not increase obviously.

The standard deviation of offsets reflects the diffusion degree of LST thermal anomalies caused by subsurface fire sources, and it decreases with time in east direction (T1: 37.40 m, T2: 32.93 m, and T3: 27.23 m). Specifically, the subsurface fire source burns intensely at T1, and the influence range of LST thermal anomaly caused by it is large, increasing the difficulty of inferring the subsurface fire source location through the LST thermal anomaly. There are three fire sources at T2, and the combustion range is larger than T1; thus, the standard deviation of offsets does not decrease too much even though coal fires burn less intensely. The offset standard deviation of T3 decreases due to its smaller burning scale. The offset standard deviation in east direction is larger than that in north direction, indicating that the LST thermal anomaly spreads more obviously along the coal seam strike. The offset standard deviation in north direction does not vary much (T1: 11.33 m, T2: 11.92 m, and T3: 10.80 m), indicating that the LST thermal anomaly has a small degree of spread along the coal seam dip. Given these points, the relative error of coal seam strike may be greater than that of coal seam dip when inferring the locations of subsurface fire sources through LST thermal anomalies.

C. Modeling of Offsets Versus Time and Fire Source Size

This section models the offset of the subsurface fire source relative to the LST thermal anomaly as a function of time and fire source size, respectively (see Fig. 17). The offset values in the eastern and vertical directions are significantly negatively correlated with time, with linear correlation coefficients R^2 of

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TABLE IV
CHANGE RATE AND CHANGE SPEED OF OFFSETS BETWEEN UNDERGROUND FIRE SOURCES AND LST THERMAL ANOMALIES

	Change rate (%)			Change speed (m/day)		
	T1 ∼ T2	T2 ∼ T3	T1 ∼ T3	T1 ∼ T2	T2 ∼ T3	T1 ∼ T3
Eastern offset	-29.10	-13.89	-38.95	-2.39	-0.81	-1.60
Northern offset	2.88	43.72	47.85	0.05	0.84	0.45
Vertical offset	-37.60	-27.73	-54.90	-1.86	-0.85	-1.36

0.9247 and 0.9564, respectively. The offset values in the northern direction are positively correlated with time, with R^2 of 0.7949 [see Fig. 17(a)]. The change rates of eastern offset values for T1∼T2 and T2∼T3 are −29.10% and −13.89%, respectively, and the change speeds are −2.39 and −0.81 m/day, respectively (see Table IV). The offset distance between the underground fire source and the LST thermal anomaly in the eastern direction gradually decreases, and the decrease rate of T2∼T3 becomes one-third of T1∼T2. The change rate of the vertical offset for T1∼T3 reaches −54.90%, and the fire source moves toward the surface at a speed of 1.36 m/day. In contrast, the offset value gradually increases in the northern direction. The change rate of offset value for T2∼T3 is 43.72%, but the change speed of offset value is smaller than that in the east direction, only 0.45 m/day in T1∼T3, and only 0.84 m/day even in T2∼T3 where the change rate of offset value is the fastest. This indicates that the offset change in the north direction is not significant.

The standard deviations of offsets reflect the dispersion degree of the LST thermal anomalies caused by underground fire sources. It is negatively correlated with time in eastern, northern, and vertical directions, which indicates a trend of aggregation of LST thermal anomalies. Fig. 17(a)–(d) shows a clear symmetry, which indicates that the averages and standard deviations of offsets show opposite characteristics with respect to time and fire source size.

VI. CONCLUSION

Precise localization of underground fire sources is critical for coalfield fire extinguishing. This study reconstructs the 4-D temperature fields of underground fire sources by the EBK3D method for the first time. Then, the feasibility of detecting small-scale concealed fire sources by SAT TIR RS and UAV TIR RS is discussed. Finally, the refined LST thermal anomaly results obtained by UAV TIR RS and the reconstructed underground fire sources are combined to reveal the spatiotemporal correlation characteristics between the LST thermal anomalies and the subsurface fire sources. The main contributions and conclusions of the study are given as follows.

1) It is feasible to reconstruct the 4-D temperature fields of the subsurface fire sources by interpolating the borehole temperature measurement data with the EBK3D method. The reconstruction results can characterize the spatial and temporal evolution of the subsurface fire sources. The cross-validated RMSEs of the reconstructed fire sources for the three time periods are, respectively, 52.2 °C, 49.6 °C, and 37.1 °C, and R^2's of linear regression are 0.925, 0.9145, and 0.8429, respectively.
2) SAT TIR RS can provide effective detection of overall LST thermal anomalies, but its ability to detect small-scale concealed fire sources is limited. Influenced by the image resolution, push-broom imaging mode, and topographic effects, the hidden subsurface fire sources in the study area are not detected by SAT TIR RS. UAV TIR RS can effectively detect LST thermal anomalies related with small-scale subsurface fire sources. However, there is a deviation by tracing the location of subsurface fire sources through the detected LST thermal anomalies using the "vertical correspondence" method. The LST thermal anomalies show a significant "spatiotemporal delay" with respect to the subsurface fire source evolution.
3) This study quantifies the offsets of subsurface fire sources relative to the LST thermal anomalies in the eastern, northern, and vertical directions. The offsets in the coal seam strike and dip directions for different time periods at depths of (T1: −44.43 m, T2: −27.72 m, and T3: −20.04 m) are (T1: 73.80 m, T2: 52.33 m, and T3: 45.06 m) and (T1: 16.79 m, T2: 17.27 m, and T3: 24.82 m), respectively. Moreover, R^2's for the linear regression model of the offset averages in three directions versus time and fire source size are (0.9247, 0.7949, and 0.9564) and (0.8739, 0.85, and 0.9152), respectively.

The "spatiotemporal delay" between subsurface fire sources and surface anomaly features poses a challenge to accurately infer the locations of subsurface fire sources. This is not only present in underground coal fires but is prevalent in other similar studies by using surface anomalies to infer subsurface phenomena. In the future, by more thoroughly investigating the spatiotemporal correlation between surface anomalies and subsurface phenomena together with higher resolution RS images and detailed geological parameters, RS technology has the potential to solve these problems.

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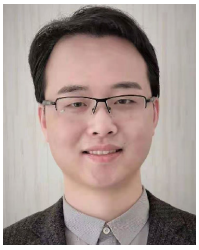
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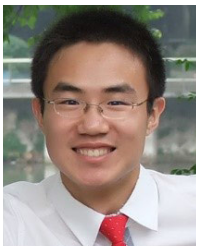
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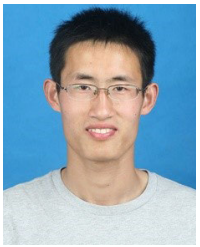
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